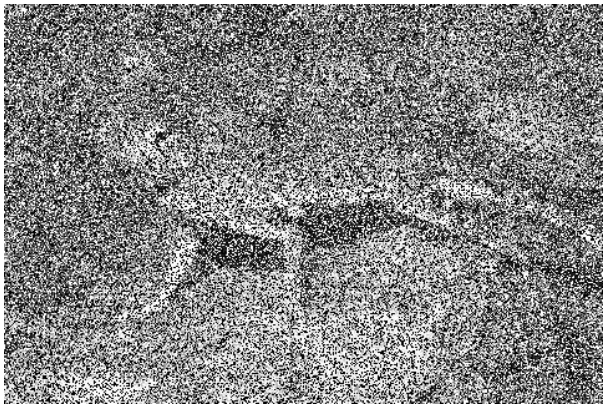
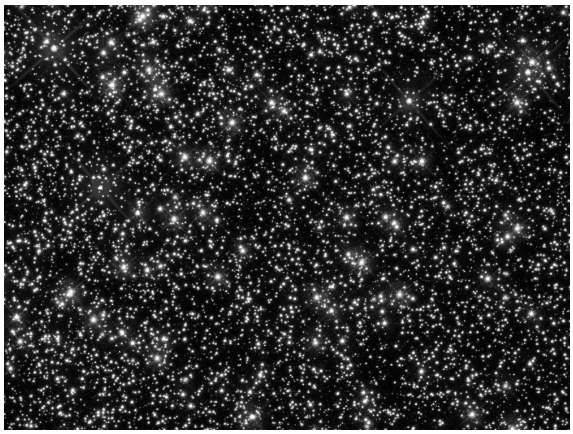
Lab 5: Image Manipulation
ISC 4221
November 4, 2021

Programming Language: Matlab or Python
Due date: 12:20PM on November 17, 2021
Submit online through Canvas
To receive full credit, you must submit:

1. Well-commented and working code
2. A single PDF writeup containing any plots, discussion, calculations, and conclusions.
3. **This lab includes a lot of plots, so it is important that they are clearly labelled in your writeup. Points will be deducted for figures that are not titled appropriately.**

Goals: The objectives of this lab are to perform contrast stretching, denoising, edge detection, and component notification. All parts of the lab will use the following five images.





1. Contrast Stretching Use the file `mona_lisa.png` for this problem. Perform the following manipulations to the image and include the manipulated image and its histogram with 256 bins.

(i) Read in the image file and display the image and its histogram

(ii) Divide the range $[0, 255]$ into three ranges with about the same number of pixels (based on the histogram). Reset pixels in the first range to 0, the middle range to 127, and the third range to 255. Display the 3-level image and its histogram with 256 bins.

(iii) Double the value of each pixel with a value in $[0, 127]$ and set all the other pixels to a value of 255. Display the brighter image and its histogram with 256 bins.

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2. Noise Removal Use the file `lizard_noisy.png` for this problem. Perform the following manipulations to the image and include a the resulting image after each step.

(i) Read in the image file and display the image.

(ii) Denoise the image by replacing each pixel with the average value of its 3×3 neighborhood.

(iii) Apply the same averaging technique a second time. Does the image improve?

(iv) Denoise the image by replacing each pixel with the median value of its 3×3 neighborhood.

(v) Apply the same median technique a second time. Does the image improve?

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3. Edge Detection Use the file `bird.png` for this problem. Implement the NEWS algorithm to detect the edges. Experiment with the level of thresholding (as we did in class) so that you can create a new picture which displays the locations of edges. Adjust the thresholding so that you achieve a "cartoon" outline of the bird. Be sure to report your thresholding parameter along with your final image.

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4. Component Identification Use the files `conjunction_gray.png` and `star_field_gray.png`.

(i) For the `conjunction_gray` file, determine a threshold parameter t between 0 and 255 so that when you threshold the image, everything is black but the two stars and the moon, which are white (255). Include a plot of your thresholded image along with your thresholding parameter.

(ii) Apply a connected component identification algorithm to your thresholded image, which returns the number of components. Ideally, there will be three components (the moon and the two stars). However, depending on your implementation, language, and threshold value, your code might interpret the moon as having three components.

(iii) For the `star_field_gray.png` file, determine a threshold parameter t between 0 and 255 so that the thresholded image separates white blobs of stars. Include a plot of your thresholded image along with your thresholding parameter.

(iv) Apply a connected component identification algorithm to your thresholded image. How many stars does your code count?